PORTER DELIVERY PERFORMANCE ANALYSIS REPORT

Prepared by: Utkarsh Dhangar

Tool: Microsoft Excel 2019 | Power Query | Pivot Tables | Pivot Charts

Dataset Size: 197,373 Orders | 15 Columns

Dataset Overview

Metric	Value
Orders	197,373
Columns	15
Avg Delivery Time	47.62 min
Avg Order Value	₹2,682.42
On-Time Rate	13.16%
Markets	7
Avg Busy Ratio	0.85

Executive Summary

Porter is a fast-growing food delivery platform facing increasing customer dissatisfaction due to delivery delays. This project analyzes operational data to identify factors influencing delivery performance and order value. The analysis focuses on delivery efficiency, market performance, restaurant categories, staffing utilization and operational consistency. Results are summarized in an interactive Excel dashboard to support data-driven decision making.

Business Problem

Customer satisfaction has declined because of longer delivery times in several markets. The objective is to identify operational bottlenecks and recommend improvements that reduce delivery time while maintaining service quality.

Project Objectives

- Analyze delivery time patterns.
- Compare market performance.
- Evaluate order protocols and staffing.
- Identify high-value restaurant categories.
- Measure operational consistency.
- Build an interactive dashboard for stakeholders.

Data Cleaning

Performed duplicate checks, missing value validation, timestamp verification, datatype corrections, and quality checks. Negative values observed in partner availability and price range were documented as data quality issues rather than removed because they represented source inconsistencies.

Feature Engineering

Created Order Hour, Order Day, Month, Weekend Flag, Busy Ratio, Price Range and Delivery Status. These derived fields enabled time-based, staffing and operational analysis.

Exploratory Data Analysis

Key analyses included:

- Order distribution by market.
- Delivery time by market and category.
- Peak ordering hours.
- Delivery time distribution.
- Average order value by market.
- Correlation between subtotal and price range.
- Correlation between order complexity and delivery time.
- Standard deviation analysis across markets and categories.
- Partner utilization analysis.
- High-value category identification.

Key Results

- Market 1 recorded the highest average delivery time (51.25 min).
- Market 4 generated the highest average order value (₹2,896).
- 130,668 deliveries occurred within 31–61 minutes.
- Peak ordering hour: 2.
- Price Range vs Subtotal correlation: 0.510 (moderate positive).
- Total Items vs Delivery Time: 0.118 (weak positive).
- Distinct Items vs Delivery Time: 0.156 (weak positive).
- Japanese category had the highest percentage of above-average orders (53.7%).
- Market 1 showed the greatest delivery variability (Std Dev 21.43).

Dashboard Overview

The dashboard contains KPI cards (Total Orders, Average Delivery Time, Average Order Value, On-Time Rate, Total Markets and Busy Ratio), market performance charts, delivery distribution, order demand by hour, market-wise order value, high-value categories and interactive slicers.

Business Insights

Delivery efficiency depends on operational execution rather than partner availability alone. High-value restaurant categories should receive focused marketing efforts. Market 1 requires immediate operational improvements, while Markets 2 and 4 demonstrate efficient performance despite high partner utilization.

Recommendations

1. Improve staffing and dispatch in Market 1.
2. Replicate best practices from Markets 2 and 4.

3. Promote Japanese, Pizza, Chinese and Indian restaurants.
4. Improve forecasting during peak demand.
5. Standardize operations for high-variance categories.
6. Improve source data quality for partner metrics.

Limitations

The dataset lacks customer identifiers, preventing repeat customer analysis and customer lifetime value calculations. Negative partner availability and pricing anomalies indicate source data quality issues.

Conclusion

This project demonstrates how Excel can be used to perform end-to-end business analytics. The dashboard converts operational data into actionable insights that help improve delivery efficiency, customer satisfaction and revenue performance.

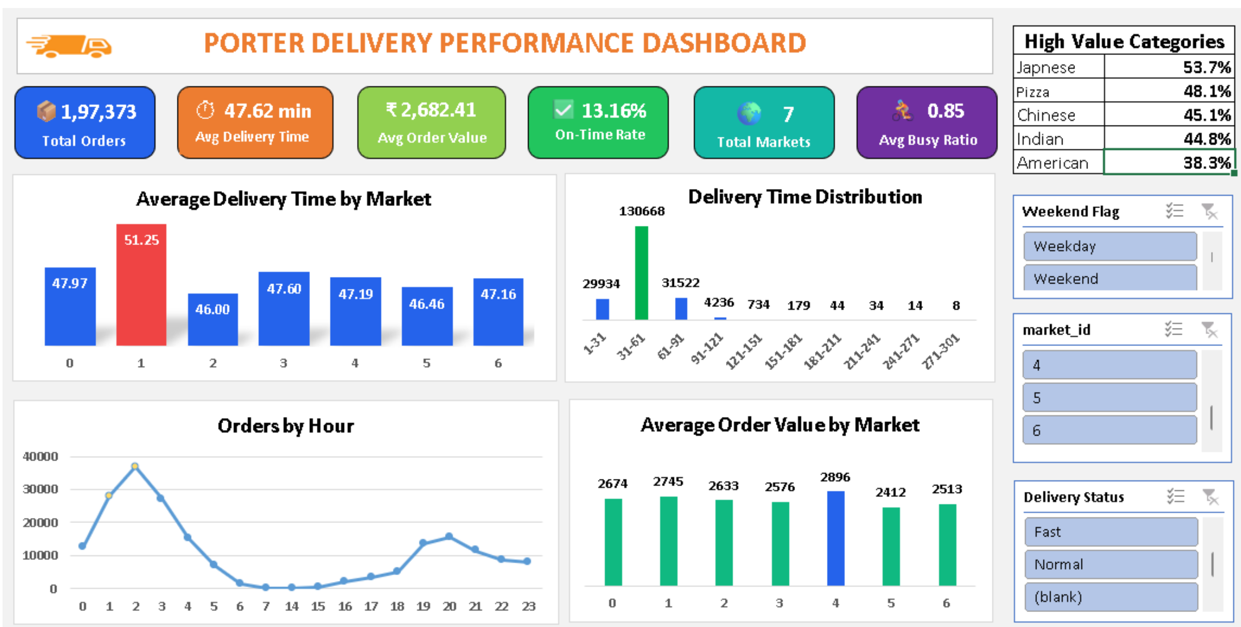

Figure: Porter Delivery Performance Dashboard